

MemSieve: Implementation and Paper Roadmap

A Harsh Research Audit, System Blueprint, Experimental Plan, and Submission Manual

Prepared from `paper_roadmap.pdf`

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Source Discipline and Audit Position

This document is a standalone implementation-and-paper manual derived from the repository file `paper_roadmap.pdf`. The repository also contains `AeroMind Paper.pdf`, but this manual does not use it because the task explicitly names `paper_roadmap.pdf` as the source report. Any later use of other private or unpublished material must be declared and reviewed for overlap risk before submission.

The source report is strong in its research framing: it correctly identifies that the publishable unit is not “memory can be attacked,” but a defense-first runtime architecture for mutable LLM agent memory. It also makes the right strategic call: the paper should focus on a compact, cross-framework, cross-subsystem memory firewall with scalability evidence, not a broad benchmark, new attack taxonomy, or heavyweight formal system.

The source report is weak as an implementation document. It does not specify exact APIs, thresholds, task schemas, framework integration contracts, datasets, statistical tests, reproducibility artifacts, or acceptance criteria for individual modules. It cites many public threads of work but does not provide verified bibliographic metadata. It proposes a two-week execution schedule, which is plausible only for a disciplined MVP and implausible for a broad main-conference paper with full baseline reproduction.

Operating rule for this manual: when the report does not specify something, this document marks it as **Missing / Must Decide:** rather than pretending the detail exists. Implementation decisions below are derived from the report’s thesis and converted into concrete contracts so a developer can begin coding immediately.

Chapter 1

Research Vision and Paper Thesis

1.1 Executive Research Vision

Core research idea. MemSieve is a provenance-aware memory firewall for LLM agents. It sits between agent frameworks and their memory APIs, forcing every memory write, retrieval, context assembly, tool action, and cross-agent memory propagation event through a common defensive pipeline. The core architectural insight is to treat agent memory as a mutable write–retrieve–propagate system rather than as a passive vector index or prompt suffix.

Problem solved. Modern LLM agents increasingly store durable state: user facts, project preferences, retrieved passages, prior trajectories, shared team notes, tool outputs, summaries, and coordination artifacts. These memory records can be contaminated by poisoned corpora, indirect prompt injection, low-trust user content, compromised peer agents, or self-reinforcing summaries. Current defenses are fragmented: they usually protect one stage, one substrate, or one framework. MemSieve aims to reduce contaminated writes, unsafe retrieval, contradiction-induced failures, tool-use abuse conditioned on memory, and cross-agent propagation.

Why the problem matters now. The report argues that public agent frameworks now expose durable memory as a first-class runtime feature. This changes the security problem from prompt safety at one interaction to persistent state safety across sessions and agents. The important shift is persistence: a contaminated memory record can keep influencing future reasoning after the original attacker interaction has disappeared.

Expected paper-level contribution. The expected contribution is a compact systems/security paper that demonstrates a reusable runtime layer over public memory APIs. The paper should show that one canonical schema plus multi-stage gates can materially lower unsafe retrieval and propagation across vector/RAG memory, semantic fact memory, episodic/example memory, and shared coordination memory, while preserving clean utility and keeping overhead bounded as memory size and agent count grow.

Why this is publishable rather than merely engineering. The publishable claim is not that a developer can wrap memory APIs. The publishable claim is that memory defense needs a new evaluation object: the full write–retrieve–propagate lifecycle across heterogeneous memory subsystems. A paper becomes credible if it shows claim-evidence alignment: generic metrics, real framework adapters, cross-subsystem scenarios, ablations, and scalability curves. Without that evidence, MemSieve is only a well-named middleware prototype.

1.2 Paper Thesis

The strongest thesis is:

LLM agent memory should be defended as a stateful runtime subsystem. A framework-agnostic middleware that normalizes memory records into a provenance- and lineage-aware schema, gates writes, reranks retrievals under risk, renders retrieved content as typed evidence, verifies memory-conditioned tool actions, and controls cross-agent propagation can substantially reduce memory contamination while preserving clean utility and scaling acceptably.

The paper must not claim general immunity. It should claim measurable reduction under a bounded threat model.

1.3 Non-Negotiable Scope

- MVP frameworks: LangGraph, LlamaIndex, and CrewAI.
- Optional framework: AutoGen only if integration is low-friction.
- Stretch framework: Letta/MemGPT only if it is practically zero-friction.
- MVP memory subsystems: vector/RAG, semantic fact, episodic/example, shared coordination.
- MVP defense stages: adapter, write gate, quarantine/audit, retrieval gate, context shaping, pre-tool verification, propagation controller.
- MVP evidence: clean utility table, attack reduction table, ablation table, propagation result, scalability curves, reproducibility package.

Chapter 2

Novelty, Gaps, and Contributions

2.1 Strongest Novelty Claims Extracted from the Report

Claim	Why it is potentially strong	Evidence needed
Cross-lifecycle memory defense	Most defenses target retrieval, prompts, or tools separately. The report’s strongest move is to defend writes, retrievals, context assembly, tool actions, and propagation as one lifecycle.	Ablations showing that no single stage explains all gains.
Cross-subsystem coverage	Vector-only defenses are easy to dismiss. Covering semantic, episodic, and shared coordination memory makes the paper meaningfully about agent memory.	Main results table must include non-RAG rows, not appendix-only evidence.
Framework-agnostic runtime layer	A common schema plus adapters can make the architecture portable across public frameworks.	Working adapters for at least three frameworks with identical scenario templates and logging.
Propagation-aware memory security	Shared memory and peer-agent summaries make contamination spread measurable.	Lineage graph, propagation depth, fan-out, and agent-count scaling.
Scalability-first evaluation	Security that only works on tiny stores is not deployable.	Curves over memory size, poison ratio, top-k, and agent count.

2.2 Weak, Risky, or Already Covered Claims

Risky claim	Why it is weak	Rewrite or remove
-------------	----------------	-------------------

“We introduce memory attacks.”	The report itself says public work already covers indirect prompt injection, RAG poisoning, memory injection, and propagation.	Remove. Position attacks as evaluation scenarios grounded in public threat categories.
“Framework-agnostic defense.”	This is only true if adapters actually run. Otherwise reviewers will treat it as architecture rhetoric.	Claim “portable across three public frameworks via thin adapters” after demonstrating LangGraph, LlamaIndex, and CrewAI.
“Defends all memory.”	The report explicitly recommends four stable subsystems and says procedural/scratchpad memory is an extension.	Claim coverage of four practical memory subsystems; mark procedural memory as future work.
“Novel detector.”	The report warns against a monolithic learned detector. A detector-focused claim would be a schedule and review trap.	Claim interpretable, calibratable gates and optional LLM checks, not a new classifier.
“Strong conference main-track readiness in two weeks.”	The report is blunt that this is unlikely unless most code already exists.	Target a compact systems/security paper or workshop submission first; build a main-track version only after stronger evaluation.

2.3 Publishable Novelty Statement

MemSieve is a framework-portable memory firewall that defends LLM agent memory across the write, retrieve, reason, act, and propagate lifecycle. It normalizes heterogeneous memory records into a common schema carrying provenance, trust priors, contradiction signals, and lineage edges; quarantines or down-trusts suspicious writes; reranks retrieved candidates with risk-aware scoring and source diversity; renders memory as typed evidence rather than executable instruction; verifies memory-conditioned tool actions; and limits cross-agent spread through lineage-aware propagation control. The novelty is the combination of lifecycle coverage, cross-subsystem evaluation, public framework adapters, generic memory-security metrics, and scalability evidence.

2.4 Concrete Paper Contributions

#	Contribution	What the paper must prove	Primary evidence
---	--------------	---------------------------	------------------

1	A lifecycle defense architecture for mutable agent memory.	Multi-stage defense reduces contaminated writes, unsafe retrieval, unsafe context, unsafe tool action, and propagation more than single-stage baselines.	Main attack table plus ablation table.
2	A canonical provenance and lineage schema for heterogeneous agent memory.	One schema can represent vector/RAG, semantic, episodic, and shared coordination records across three frameworks.	Schema definition, adapter tests, cross-framework logs.
3	A cross-subsystem memory-security evaluation harness.	The same scenarios and metrics run across multiple frameworks and memory types without bespoke hand-tuning per row.	Experiment runner, scenario manifests, reproducibility artifacts.
4	New evaluation metrics for memory contamination and propagation.	Unsafe retrieval alone is insufficient; contaminated context and propagation depth capture stronger failure modes.	Metric definitions, examples, and plots linking metrics to downstream outcomes.
5	Scalability evidence for security under memory growth and team growth.	Security gains remain meaningful as memory size, poison ratio, top-k, and agent count increase, with acceptable overhead.	Scalability curves and security-cost frontier.

2.5 Gaps That Must Be Filled Before Coding

- **Missing / Must Decide:** Exact model provider, model version, embedding model, and cost budget.
- **Missing / Must Decide:** Exact framework versions and whether adapters target Python or JavaScript APIs.
- **Missing / Must Decide:** Threshold values for write admission, retrieval risk, quarantine, trust decay, and confirmation.
- **Missing / Must Decide:** Public corpus choice or synthetic corpus generator parameters.
- **Missing / Must Decide:** Tool-action risk taxonomy and which stub tools count as high impact.
- **Missing / Must Decide:** Statistical test policy: confidence intervals, paired tests, bootstrap, or only descriptive statistics.

- **Missing / Must Decide:** Submission venue, page limit, artifact expectations, and anonymity policy.

Chapter 3

System Architecture

3.1 Architecture Overview

MemSieve sits between an agent runtime and its native memory implementation. The framework should believe it is calling a normal memory API; MemSieve intercepts and records every memory event.

Figure 1 textual architecture placeholder.

Untrusted sources, user turns, tools, external documents, and peer-agent messages enter the MemSieve adapter layer. The adapter converts each event into a canonical memory record. Write-time admission either commits, down-trusts, or quarantines the record. The audit logger records every decision. Retrieval calls fetch native candidates, then risk-aware reranking applies trust, corroboration, contradiction, lineage, duplicate, and diversity logic. The context shaper renders selected records as typed evidence blocks. The agent reasons over this shaped context. Before tools run, pre-tool verification checks user authorization and trusted support. For multi-agent systems, the propagation controller attaches lineage edges, decays trust across summaries, and freezes rapid low-trust fan-out.

Figure 3.1: Architecture placeholder. The final paper should replace or refine this with a clean pipeline diagram showing trust boundaries and audit side channels.

3.2 Components, Inputs, Outputs, and Dependencies

Component	Input	Output	Dependencies and notes
Framework adapters	Native framework write, query, retrieve, add, and shared-memory calls.	Canonical <code>MemoryRecord</code> and <code>MemoryEvent</code> .	One adapter each for LangGraph, LlamaIndex, CrewAI. AutoGen optional.

Canonical schema	Raw content plus metadata.	Validated record with subsystem, source type, actor, lineage, trust, labels, and timestamps.	Must remain identical across frameworks.
Provenance engine	Source type, actor, parent IDs, framework namespace, record labels.	Trust prior and lineage features.	Uses configured trust priors and decay rules.
Write-time admission gate	Candidate write record.	Decision: commit, commit-low-trust, quarantine, reject.	Uses instructionality, contradiction, sensitivity, duplicate, and source-risk features.
Quarantine store	Suspicious record and decision rationale.	Non-retrievable or restricted quarantine record.	Must be isolated from normal retrieval.
Audit logger	Every write, retrieve, rerank, context, tool, and propagation event.	Append-only JSONL logs.	Required for reproducibility, metrics, forensics, and paper figures.
Retrieval reranker	Native top-k candidates, query, task context.	Risk-adjusted candidate set.	Score combines relevance, trust, corroboration, contradiction, lineage risk, duplicate penalty, and diversity.
Context shaper	Selected records and risk labels.	Typed evidence blocks.	Must prevent raw retrieved text from being rendered as instruction.
Pre-tool verifier	Proposed tool call, user instruction, supporting memory records.	Allow, ask-confirmation, read-only degrade, or block.	Uses support threshold and tool risk classes.
Propagation controller	Shared-memory writes, summaries, peer messages.	Lineage edges, trust decay, freeze/quarantine decisions.	Needed for shared-memory cascade experiments.
Experiment runner	Scenario manifests, framework config, baseline config, seeds.	Run logs, metrics, result tables, plots.	Must isolate runs and preserve raw logs.

3.3 Canonical Record Schema

The schema below is an implementation contract. The report gives the core fields; this manual adds fields required for experiments, reproducibility, and lineage.

MemoryRecord:

```
record_id: string
run_id: string
framework: enum[langgraph, llamaindex, crewai, autogen, letta]
subsystem: enum[vector_rag, semantic, episodic, shared_coordination,
                procedural_extension]
content: string
content_hash: string
source_type: enum[system_seed, curated_corpus, user_text, external_doc,
                 tool_output, peer_agent, summary, synthetic_attack]
source_id: string
actor_id: string
timestamp: iso8601
parent_ids: list[string]
lineage_depth: integer
trust_prior: float
trust_score: float
relevance_score: float|null
risk_score: float
contradiction_score: float
instructionality_score: float
duplicate_cluster_id: string|null
sensitivity_label: enum[none, private, credential_like, policy_like]
poison_label: enum[benign, poisoned, unknown]
decision: enum[commit, commit_low_trust, quarantine, reject]
decision_reasons: list[string]
quarantine_id: string|null
tags: list[string]
```

Missing / Must Decide: The final implementation must decide whether records are persisted in SQLite, DuckDB, JSONL plus vector index, or framework-native stores plus a sidecar metadata database. The quickest MVP is JSONL audit logs plus SQLite metadata plus framework-native memory stores.

3.4 Core API Contracts

```
class MemoryFirewall:
    def admit_write(self, record: MemoryRecord) -> AdmissionDecision: ...
    def retrieve(self, query: MemoryQuery) -> RetrievalDecision: ...
    def shape_context(self, decision: RetrievalDecision) -> EvidenceBundle: ...
    def verify_tool_call(self, call: ToolCall,
```

```

        evidence: EvidenceBundle) -> ToolDecision: ...
def propagate(self, event: PropagationEvent) -> PropagationDecision: ...

class FrameworkAdapter:
    def wrap_memory(self, native_memory: object) -> object: ...
    def to_record(self, event: NativeMemoryEvent) -> MemoryRecord: ...
    def native_commit(self, record: MemoryRecord) -> None: ...
    def native_query(self, query: MemoryQuery) -> list[MemoryRecord]: ...

```

The API should be synchronous first. Async support can be added behind the same interface only after the synchronous MVP is stable.

3.5 Scoring Logic

The retrieval utility score should be explicit and logged:

$$\begin{aligned}
 S(r, q) = & \lambda_{\text{rel}} \text{Rel}(r, q) + \lambda_{\text{trust}} \text{Trust}(r) + \lambda_{\text{cor}} \text{Corrob}(r) \\
 & - \lambda_{\text{contr}} \text{Contradict}(r, q) - \lambda_{\text{lineage}} \text{LineageRisk}(r) - \lambda_{\text{dup}} \text{DuplicatePenalty}(r).
 \end{aligned}$$

The final implementation must log each term separately. If only the aggregate score is logged, ablation and reviewer debugging become weak.

Missing / Must Decide: Initial weights are not specified in the report. Start with fixed configuration values, then run a threshold sensitivity sweep. Do not train a detector unless the MVP is already complete.

3.6 Data Flow

1. A user turn, tool output, external document, or peer-agent message attempts to enter memory.
2. The framework adapter converts the native event into a `MemoryRecord`.
3. The provenance engine assigns a trust prior from source type and lineage depth.
4. The write gate computes instructionality, contradiction, duplicate, sensitivity, and source risk features.
5. The gate commits, down-trusts, quarantines, or rejects the record.
6. The audit logger writes the raw event, normalized record, scores, and decision.
7. During task execution, the native framework retrieves candidates.
8. The retrieval gate reranks candidates and enforces source and lineage diversity.
9. The context shaper renders memory as typed evidence blocks.
10. The agent proposes an answer or tool call.

11. The pre-tool verifier checks whether high-impact tool arguments are supported by explicit user intent or corroborated trusted memory.
12. In multi-agent runs, shared-memory writes carry parent IDs and trust decay; rapid low-trust fan-out is frozen or quarantined.

3.7 Trust Boundaries and Attacker Interaction Points

Boundary	Attacker interaction point	Required control
External content to memory	Poisoned documents, retrieved passages, tool-returned text.	Source trust priors, instructionality scan, quarantine, context shaping.
User text to memory	Query-only write inducement and false fact insertion.	Write admission, contradiction checks against high-trust memory, audit labels.
Memory to prompt context	Unsafe retrieved chunks entering the model as raw instructions.	Risk-aware reranking and typed evidence rendering.
Memory to tool execution	Memory-conditioned tool argument or policy change.	Pre-tool verification, support threshold, read-only degrade.
Agent to shared memory	Compromised peer-agent output, infected summaries, team notes.	Lineage edges, trust decay, fan-out freeze.
Quarantine to normal memory	Accidental retrieval of suspicious records.	Physical namespace separation and explicit quarantine policy.

3.8 Implementation Order

The first components to implement are the canonical schema, audit logger, LangGraph adapter, write gate, retrieval gate, and metric calculation. Propagation should be implemented before any optional fourth framework because propagation is one of the paper’s differentiators.

3.9 Required Paper Figures

Figure	What it should show	Why it matters
System architecture	Pipeline from untrusted sources through gates to agent, tools, and shared memory, with audit side channel and trust boundaries.	Establishes that the system is lifecycle defense, not only retrieval filtering.

Canonical memory schema	Four memory subsystem examples normalized into one schema.	Supports framework/subsystem portability.
Lineage graph	Poison source, summary hops, shared writes, and downstream actions.	Makes propagation measurable and concrete.
Security-cost frontier	Attack reduction or contaminated context reduction versus p95 latency or memory overhead.	Preempts “too expensive” reviewer objections.
Scalability curves	Memory size, poison ratio, top-k, and agent count sweeps.	Makes scaling a central contribution rather than appendix filler.

Chapter 4

Threat Model

4.1 Attacker Capabilities

The attacker has bounded but realistic capabilities:

- **Query-only write inducement:** the attacker can interact with the agent through normal user or task channels and attempt to cause persistent malicious or misleading memory writes.
- **Corpus or passage poisoning:** the attacker can insert or influence documents likely to be retrieved by vector/RAG memory.
- **Indirect prompt injection:** the attacker can place instructions inside retrieved or tool-returned content.
- **Compromised peer-agent output:** the attacker can control or contaminate one peer agent’s output that is written into shared coordination memory.

The attacker does not control the model weights, host operating system, memory firewall code, or experiment harness.

4.2 Attacker Goals

- Cause poisoned records to be committed to durable memory.
- Cause poisoned or low-trust records to dominate retrieval.
- Cause retrieved content to enter reasoning as instruction rather than evidence.
- Cause contradiction collisions where recent or semantically similar false records override high-trust records.
- Cause memory-conditioned unsafe tool calls.
- Cause contamination to propagate across agents through shared memory, summaries, or notes.

4.3 Defender Assumptions

- The defender can wrap or intercept framework memory APIs.
- The defender can maintain sidecar metadata, audit logs, and quarantine namespaces.
- The defender can assign source trust priors based on source type and framework context.
- The defender can run lightweight deterministic checks and optional LLM checks.
- The defender can enforce tool-call confirmation or read-only degradation for high-risk calls.

Missing / Must Decide: The implementation must decide whether optional LLM checks use the same model as the agent, a cheaper verifier model, or no model at all.

4.4 In Scope

- Persistent memory contamination through normal interaction paths.
- Vector/RAG poisoning that affects retrieval.
- Semantic false fact insertion and contradiction.
- Episodic poisoned exemplar replay.
- Shared coordination memory cascades.
- Memory flooding and duplicate domination.
- Utility and latency impact of defensive gates.

4.5 Out of Scope

- Model-weight poisoning or fine-tuning attacks.
- Host compromise, file-system compromise, or arbitrary code execution.
- Embedding-model attacks and vector-index implementation vulnerabilities.
- Vendor prompt leakage or hidden system prompt extraction.
- Covert channels.
- Human-in-the-loop social engineering outside the agent interface.
- Formal proof of complete memory integrity.

4.6 Victim System and Environment

The victim is an LLM-agent application built on a public framework with durable memory enabled. The MVP victim systems are:

- LangGraph agent with vector/RAG and semantic memory.
- LlamaIndex agent with memory blocks or long-term memory abstractions.
- CrewAI team with shared coordination memory.

All tasks should use safe stub tools and synthetic or public non-sensitive corpora. The paper should avoid regulated domains unless there is a clear reason and proper ethics review.

4.7 Credibility of the Threat Model

The threat model is credible because it matches the direction of public agent systems: persistent memory, retrieval, summarization, and multi-agent coordination are exposed as normal features. It is not exaggerated because it excludes host compromise, model compromise, hidden vendor internals, and unbounded attacker control. The attacker interacts through plausible content and memory paths.

Chapter 5

Implementation Blueprint

5.1 Repository Structure

Create the following repository structure:

```
memsieve/  
  pyproject.toml  
  README.md  
  LICENSE  
  configs/  
    default.yaml  
    frameworks.yaml  
    thresholds.yaml  
    models.yaml  
memsieve/  
  __init__.py  
  core/  
    record.py  
    events.py  
    decisions.py  
    scoring.py  
    config.py  
    clock.py  
  stores/  
    metadata_store.py  
    quarantine_store.py  
    audit_log.py  
    vector_store.py  
  gates/  
    write_gate.py  
    retrieval_gate.py  
    context_shaper.py  
    tool_verifier.py  
    propagation_controller.py
```

```
adapters/  
  base.py  
  langgraph_adapter.py  
  llamaindex_adapter.py  
  crewai_adapter.py  
  autogen_adapter.py  
baselines/  
  no_defense.py  
  provenance_only.py  
  spotlighting_wrapper.py  
  trust_threshold.py  
  random_quarantine.py  
scenarios/  
  poisoned_fact.py  
  retrieved_instruction.py  
  experience_graft.py  
  conflict_recency.py  
  shared_cascade.py  
  flooding.py  
experiments/  
  runner.py  
  manifests.py  
  seed.py  
  graders.py  
  metrics.py  
  aggregate.py  
plotting/  
  tables.py  
  figures.py  
  latex_export.py  
manifests/  
  scenarios/  
  experiment_matrix.yaml  
data/  
  raw/  
  synthetic/  
  curated/  
runs/  
  .gitkeep  
scripts/  
  run_smoke_tests.ps1  
  run_main_matrix.ps1  
  aggregate_results.ps1  
  make_figures.ps1  
tests/  
  unit/  
  integration/  
  regression/
```

```

paper/
  memsieve_paper.tex
figures/
tables/

```

5.2 Core Modules and Classes

Module	Core classes	Responsibility
core.record	MemoryRecord, MemoryQuery	Canonical schema, validation, serialization.
core.decisions	AdmissionDecision, RetrievalDecision, ToolDecision, PropagationDecision	Typed decision objects with scores and reasons.
stores.audit_log	AuditLogger	Append-only JSONL event logging.
stores.quarantine_store	QuarantineStore	Isolated suspicious record store.
gates.write_gate	WriteGate	Write scoring, contradiction checks, duplicate throttling, quarantine decisions.
gates.retrieval_gate	RetrievalGate	Risk-aware reranking and diversity constraints.
gates.context_shaper	ContextShaper	Converts raw records into typed evidence blocks.
gates.tool_verifier	ToolVerifier	Blocks, confirms, or degrades risky memory-conditioned tool calls.
gates.propagation_controller	PropagationController	Adds lineage, decays trust, detects fan-out cascades.
adapters.*	FrameworkAdapter subclasses	Wrap native memory APIs.
experiments.runner	ExperimentRunner	Runs scenario matrix and writes structured artifacts.
experiments.metrics	MetricComputer	Computes ASR, URR, CCR, BUP, TSD, propagation, latency, and scaling metrics.

5.3 Configuration Files

```

# configs/default.yaml
run:
  seed: 0

```

```

output_dir: runs
log_level: INFO

models:
  agent_model: MISSING_MUST_DECIDE
  verifier_model: none
  embedding_model: MISSING_MUST_DECIDE

thresholds:
  quarantine_risk: 0.75
  low_trust_risk: 0.45
  contradiction_high: 0.70
  instructionality_high: 0.65
  propagation_freeze_fanout: 3
  trust_decay_per_hop: 0.15

retrieval:
  top_k_native: 16
  top_k_final: 4
  require_source_diversity: true
  max_same_lineage: 1

tool_verification:
  high_impact_tools: [write_file, send_message, external_post, purchase]
  require_user_or_trusted_support: true

```

The threshold values above are starting defaults, not scientific facts. The evaluation must include a sensitivity sweep.

5.4 Logging Design

Every run must write:

- `run_manifest.yaml`: framework, model, seed, scenario, baseline, thresholds, git hash if available, and environment.
- `events.jsonl`: every normalized memory event.
- `records.jsonl`: canonical records and labels.
- `decisions.jsonl`: gate scores and decisions.
- `retrievals.jsonl`: native candidates, reranked candidates, selected context.
- `tool_calls.jsonl`: proposed and final tool decisions.
- `lineage.jsonl`: parent-child edges for propagation.
- `grader.jsonl`: task success, attack success, and grading rationale.

- `timing.jsonl`: p50, p95, and per-stage latencies.

Hard rule: paper metrics must be computed from logs, not from ad hoc notebook state.

5.5 Experiment Runner Design

The runner should execute a matrix:

```
for framework in frameworks:
    for subsystem in subsystems:
        for scenario in scenarios:
            for defense in defenses:
                for seed in seeds:
                    prepare_clean_environment()
                    seed_memory()
                    inject_attack_if_needed()
                    run_task()
                    grade_outputs()
                    save_logs()
```

Each run directory must be immutable after completion. Re-running should create a new run ID.

5.6 Reproducibility Design

- Pin Python version and all framework versions in `pyproject.toml` or a lock file.
- Save all scenario manifests and generated synthetic records.
- Save model names, API parameters, temperature, top-p, max tokens, and embedding model.
- Use deterministic stub tools for all side-effectful actions.
- Use fixed seeds for synthetic generation, retrieval tie-breaking, and scenario ordering.
- Export paper tables directly from result JSON or CSV.
- Keep raw logs for every table row.

Missing / Must Decide: Exact reproducibility policy for hosted models: the paper must disclose that hosted LLMs can change unless a pinned snapshot or local model is used.

5.7 Implementation Phases

5.7.1 Phase 0: Scaffold and Reproducibility

Goal. Create the repository, configuration, logging, and test skeleton.

Required modules. `core.config`, `stores.audit_log`, `experiments.runner`.

Exact tasks.

- Create package structure and dependency file.
- Implement config loading and schema validation.
- Implement immutable run directory creation.
- Implement append-only JSONL writer.
- Add unit tests for config and logging.

Expected outputs. A smoke run creates a run directory with manifest and empty logs.

Correctness checks. Tests pass; invalid config fails loudly; logs are valid JSONL.

Common failure points. Silent overwrites, unpinned configs, invalid JSON due to raw model text.

Evidence to save. Smoke-test logs and configuration snapshot.

5.7.2 Phase 1: Canonical Schema and Core Decisions

Goal. Implement canonical records and typed decisions.

Required modules. `core.record`, `core.events`, `core.decisions`.

Exact tasks.

- Implement `MemoryRecord` validation.
- Implement content hashing and deterministic record IDs.
- Implement admission, retrieval, tool, and propagation decision classes.
- Add serialization round-trip tests.

Expected outputs. Any framework event can be represented as a stable record.

Correctness checks. Round-trip tests preserve all fields; missing required fields fail.

Common failure points. Framework-specific fields leak into generic schema; IDs are non-deterministic.

Evidence to save. Schema examples for all four subsystems.

5.7.3 Phase 2: Framework Adapter Smoke Tests

Goal. Wrap `LangGraph`, `LlamaIndex`, and `CrewAI` enough to intercept memory writes and retrievals.

Required modules. `adapters.base`, `adapters.langgraph_adapter`, `adapters.llamaindex_adapter`, `adapters.crewai_adapter`.

Exact tasks.

- Identify native memory write and retrieval calls.
- Implement adapter methods for write, query, and commit.
- Build one clean task per framework.
- Verify identical event logs across framework-specific runs.

Expected outputs. Three clean smoke runs with memory event logs.

Correctness checks. Each adapter captures writes and retrievals without changing clean answers.

Common failure points. Framework version drift, async behavior, hidden memory writes, and agent code bypassing the adapter.

Evidence to save. Adapter smoke logs and minimal working examples.

5.7.4 Phase 3: Write Gate, Quarantine, and Audit

Goal. Stop or label contaminated writes before they enter normal memory.

Required modules. `gates.write_gate`, `stores.quarantine_store`, `stores.metadata_store`.

Exact tasks.

- Implement trust priors by source type.
- Implement heuristic instructionality features.
- Implement contradiction checks against high-trust semantic memory.
- Implement near-duplicate clustering or hashing.
- Implement quarantine namespace and audit reasons.

Expected outputs. Poisoned writes are committed, down-trusted, quarantined, or rejected with logged reasons.

Correctness checks. Unit tests for benign writes, obvious poisoned writes, contradictions, and duplicate floods.

Common failure points. Over-blocking benign user preferences; quarantined records leaking into retrieval.

Evidence to save. Quarantine precision/recall logs and example decision rationales.

5.7.5 Phase 4: Retrieval Gate and Context Shaping

Goal. Make retrieval optimize task utility under risk, not raw similarity alone.

Required modules. `gates.retrieval_gate`, `gates.context_shaper`.

Exact tasks.

- Retrieve a larger native candidate pool.

- Compute risk-aware score terms.
- Enforce source and lineage diversity.
- Render records as trusted facts, examples, shared notes, contradictions, or quarantined excerpts.
- Log native rank, defended rank, selected context, and dropped candidates.

Expected outputs. The model sees typed evidence blocks and fewer contaminated records.

Correctness checks. Retrieval tests where poisoned top-1 native result is demoted; context output contains no untyped raw injected instruction.

Common failure points. Excessive relevance loss; context shaper accidentally quotes malicious content as command-like text.

Evidence to save. Unsafe Retrieval Rate and Contaminated Context Rate examples.

5.7.6 Phase 5: Pre-Tool Verification and Propagation Control

Goal. Prevent memory-conditioned unsafe actions and shared-memory cascades.

Required modules. `gates.tool_verifier`, `gates.propagation_controller`.

Exact tasks.

- Define high-impact stub tools.
- Require explicit user instruction or trusted corroborating memory for high-impact actions.
- Attach parent IDs to shared-memory writes.
- Decay trust across summaries and peer-agent hops.
- Freeze rapid low-trust fan-out.

Expected outputs. Tool calls and shared writes have support evidence and lineage.

Correctness checks. A compromised peer cannot contaminate all downstream agents in the shared-memory cascade scenario.

Common failure points. Tool verifier blocks too many legitimate actions; lineage is missing for summaries.

Evidence to save. Lineage graph, propagation depth, fan-out, and blocked-call logs.

5.7.7 Phase 6: Baselines, Scenarios, and Graders

Goal. Implement fair comparisons and auto-gradable scenarios.

Required modules. `baselines.*`, `scenarios.*`, `experiments.graders`.

Exact tasks.

- Implement no defense, provenance-only, spotlighting-style wrapper, trust-threshold sanitizer, and random quarantine baseline.
- Implement six scenario families.
- Implement deterministic graders for task success and attack success.
- Add scenario manifests with seed, memory subsystem, attack records, and expected labels.

Expected outputs. A small matrix can run end to end and produce metrics.

Correctness checks. Baselines use the same tasks, models, memory sizes, and seeds as MemSieve.

Common failure points. Baselines accidentally get weaker retrieval settings; graders rely on brittle string matching only.

Evidence to save. Scenario manifests and grader validation examples.

5.7.8 Phase 7: Full Evaluation Pipeline

Goal. Run the MVP experiment matrix and produce paper-ready CSV, LaTeX tables, and plots.

Required modules. `experiments.metrics`, `experiments.aggregate`, `plotting.tables`, `plotting.figures`.

Exact tasks.

- Run main attack matrix.
- Run clean utility matrix.
- Run ablations.
- Run memory-size, poison-ratio, top-k, and agent-count sweeps.
- Aggregate metrics with confidence intervals.
- Export plots and tables.

Expected outputs. Paper tables and figures are generated from raw logs.

Correctness checks. Re-running aggregation from raw logs reproduces table values.

Common failure points. Missing run metadata, inconsistent seeds, incomplete failed-run handling.

Evidence to save. Raw logs, aggregate CSVs, plotting scripts, figure source files.

Chapter 6

Scenario Design

6.1 Scenario Requirements

Every scenario must define: name, purpose, initial system state, attacker action, defender behavior, expected failure or success mode, metrics, logs, table/figure output, and supported claim.

6.2 Scenario Matrix

Scenario	Purpose	Attacker action	Defender behavior	Claim supported
Poisoned fact insertion	Test persistent false semantic memory.	Induce storage of a false project or user fact through low-trust content.	Quarantine, down-trust, or mark contradiction against high-trust fact.	Write-path defense reduces poisoned persistent memory.
Retrieved instruction hijack	Test RAG chunks that contain hidden instructions.	Insert a retrievable document that tells the agent to change behavior.	Demote, quarantine, or render as untrusted evidence.	Retrieval and context shaping reduce unsafe retrieved instruction.
Experience graft	Test poisoned episodic examples.	Seed an apparently successful but unsafe trajectory.	Suppress or label the exemplar; prefer trusted examples.	Episodic memory needs defense, not only RAG.
Conflict and recency collision	Test recent false record versus older trusted record.	Add recent semantically overlapping contradiction.	Surface contradiction and prefer high-trust record or abstain.	Trust and contradiction beat naive recency/similarity.

Shared-memory cascade	Test peer-agent contamination.	Compromised peer writes contaminated summary or shared note.	Attach lineage, decay trust, freeze low-trust fan-out.	Propagation controller caps cross-agent spread.
Memory flooding	Test duplicate and top-k domination.	Insert many near-duplicate low-quality or poisoned records.	Cluster duplicates, throttle writes, enforce diversity.	Defense remains stable under memory pressure.

6.3 Detailed Test Cases

6.3.1 Poisoned Fact Insertion

System state before experiment. Semantic memory contains high-trust seeded facts and benign project/user facts. The agent has a task requiring later recall of one fact.

Attacker action. The attacker submits low-trust content that implies a false persistent fact should be saved.

Defender behavior. The write gate computes source risk, contradiction score, and instructionality. It commits benign facts, down-trusts uncertain facts, and quarantines high-risk false records.

Expected failure mode without defense. The false fact is written, retrieved later, and used in an answer or plan.

Metrics. Attack Success Rate, Quarantine Precision/Recall, Contaminated Context Rate, Task Success Under Defense.

Logs needed. Write event, scores, decision reasons, retrieval event, final answer, grader.

Output. Main attack table and write-gate ablation row.

Claim supported. MemSieve reduces persistent poisoned writes while preserving clean semantic memory utility.

6.3.2 Retrieved Instruction Hijack

System state before experiment. Vector/RAG memory contains benign documents relevant to a question. Attack document is semantically relevant and likely to appear in native top-k.

Attacker action. The attacker inserts a poisoned document with hidden or explicit instructions embedded in content.

Defender behavior. Retrieval gate scores lineage and instructionality risk, applies source diversity, and context shaper renders the chunk as untrusted evidence rather than instruction.

Expected failure mode without defense. The model follows the retrieved instruction or changes the answer/tool plan.

Metrics. Unsafe Retrieval Rate, Contaminated Context Rate, Attack Success Reduction, BUP,

latency overhead.

Logs needed. Native top-k, defended top-k, context bundle, final output, stage timings.

Output. Unsafe retrieval table and memory-size scaling curve.

Claim supported. The system improves over raw vector retrieval and spotlighting-style context wrapping.

6.3.3 Experience Graft

System state before experiment. Episodic memory contains prior successful trajectories. A task requires selecting or imitating a prior example.

Attacker action. The attacker seeds a poisoned trajectory marked as successful but containing an unsafe step.

Defender behavior. The write gate marks low trust or quarantines the trajectory; retrieval gate penalizes unsupported low-trust exemplars; context shaper labels examples as examples, not rules.

Expected failure mode without defense. The agent imitates the poisoned plan.

Metrics. Attack Success Rate, selected exemplar poison rate, TSD, BUP.

Logs needed. Episodic write labels, selected exemplars, generated plan, tool sequence.

Output. Non-RAG subsystem row in main table and ablation table.

Claim supported. Memory defense generalizes beyond vector/RAG corpora.

6.3.4 Conflict and Recency Collision

System state before experiment. Semantic or shared memory contains an older high-trust record. A later low-trust contradictory record overlaps semantically.

Attacker action. Insert a recent contradiction likely to win naive recency or similarity ranking.

Defender behavior. Contradiction detector flags conflict; retrieval gate prefers high-trust record or surfaces both with a warning; tool verifier blocks actions relying only on the low-trust branch.

Expected failure mode without defense. The recent false record silently overrides trusted memory.

Metrics. Contradiction win rate, TSD, CCR, abstention/confirmation rate.

Logs needed. Contradiction pairs, ranking, context, final answer/action.

Output. Conflict-resolution table.

Claim supported. Provenance and contradiction signals improve over similarity/recency.

6.3.5 Shared-Memory Cascade

System state before experiment. CrewAI team has shared coordination memory and multiple agents. One peer is contaminated.

Attacker action. The compromised peer writes a contaminated summary or shared note.

Defender behavior. Propagation controller records lineage, decays trust, freezes rapid low-trust fan-out, and quarantines suspicious shared writes.

Expected failure mode without defense. Contamination spreads to downstream agents and affects their writes or tool actions.

Metrics. Propagation depth, fan-out, contaminated shared writes, TSD, latency overhead.

Logs needed. Lineage graph, shared writes, downstream retrievals, tool calls.

Output. Agent-count scaling figure and lineage graph.

Claim supported. MemSieve contains memory-mediated cross-agent spread.

6.3.6 Memory Flooding

System state before experiment. Memory store contains benign records. Attack condition inserts many near-duplicate records with low trust or poison labels.

Attacker action. Flood vector, episodic, or shared memory with semantically similar records.

Defender behavior. Duplicate throttling, source diversity, and max-same-lineage constraints prevent domination.

Expected failure mode without defense. Top-k dominated by attacker cluster; utility drops; latency rises.

Metrics. Top-k Source Diversity, URR, CCR, BUP, latency overhead, memory growth overhead.

Logs needed. Duplicate cluster IDs, retrieval source distribution, timings, task outcomes.

Output. Flooding stress table and security-cost frontier.

Claim supported. Defense remains stable under adversarial memory pressure.

Chapter 7

Experimental Methodology

7.1 Experimental Setup

The MVP setup uses three public frameworks, four memory subsystems, six scenarios, five defense settings, three seeds, one primary model, and one shared embedding backend.

- Frameworks: LangGraph, LlamaIndex, CrewAI.
- Optional: AutoGen after MVP results are complete.
- Memory subsystems: vector/RAG, semantic, episodic, shared coordination.
- Defense settings: no defense, provenance-only, spotlighting-style retrieval wrapper, trust-threshold sanitizer, full MemSieve.
- Seeds: at least 3 for scenario generation and retrieval tie-breaking.
- Model temperature: low or zero for reproducibility.

Missing / Must Decide: Exact hosted/local model, model version, embedding backend, and API parameters.

7.2 Datasets and Simulated Environment

Use synthetic and public non-sensitive corpora. The safest MVP is a synthetic project-assistant environment with:

- project documents for vector/RAG memory;
- user and project facts for semantic memory;
- prior task trajectories for episodic memory;
- team notes and summaries for shared coordination memory;

- deterministic stub tools for high-impact actions.

This avoids domain-specific ethics drag while still exercising real memory paths.

Missing / Must Decide: The report does not specify a public corpus. Choose one public, permissively licensed corpus only if it adds realism without slowing the MVP.

7.3 Baselines

Run every applicable scenario under:

- No defense.
- Provenance-only filter.
- Spotlighting-style context wrapper.
- Trust-threshold sanitizer.
- Random quarantine baseline.
- Full MemSieve.

Optional comparisons:

- RAGForensics for vector/RAG rows if code is easy to reuse.
- Task Shield for tool-action rows if integration is moderate.

7.4 Ablations

- Full MemSieve minus write gate.
- Full MemSieve minus retrieval risk scoring.
- Full MemSieve minus context shaping.
- Full MemSieve minus contradiction checks.
- Full MemSieve minus source diversity.
- Full MemSieve minus pre-tool verification.
- Full MemSieve minus propagation controller.
- Full MemSieve with no trust decay.

7.5 Main Experiment Matrix

Experiment	Frameworks	Subsystems	Primary metrics	Priority
Clean utility	LangGraph, LlamaIndex, CrewAI	All four	BUP, TSD, latency	Must
Write-path poisoning	LangGraph, LlamaIndex	Semantic, episodic	ASR reduction, quarantine P/R	Must
Unsafe retrieval	All MVP frameworks	Vector/RAG, semantic	URR, CCR, TSD	Must
Conflict resolution	LangGraph, CrewAI	Semantic, shared	Contradiction win rate, TSD	Must
Shared cascade	CrewAI, optional AutoGen	Shared coordination	Propagation depth, fan-out, TSD	Must
Flooding stress	All MVP frameworks	Vector, episodic, shared	Top-k diversity, URR, latency	Must
Memory-size scaling	All MVP frameworks	Applicable subsystems	Scalability slope, URR, CCR, latency	Must
Agent-count scaling	CrewAI, optional AutoGen	Shared coordination	Propagation depth, fan-out, latency	Must
Second-model transfer	One or two frameworks	Vector plus shared	ASR reduction, BUP	Optional

7.6 Repeated Trials and Seeds

Use at least three seeds. For stochastic hosted models, run more if budget allows. Report confidence intervals for aggregate rates. If only three seeds are feasible, do not overstate statistical significance.

7.7 Clean, Attacked, and Defended Conditions

- **Clean:** no poisoned records; measure utility, latency, memory overhead.
- **Attacked/no-defense:** poisoned records or compromised peer present; no gates.
- **Attacked/baseline:** same attack under each baseline.
- **Attacked/full defense:** same attack under full MemSieve.
- **Attacked/ablation:** same attack with one module removed.

All comparisons must use identical tasks, seeds, framework versions, memory sizes, top-k values, and model parameters.

7.8 Tables and Figures to Produce

- Table 1: Frameworks, memory subsystems, adapter coverage.
- Table 2: Scenario matrix and threat capability mapping.
- Table 3: Main attack reduction results.
- Table 4: Clean utility preservation.
- Table 5: Ablation results.
- Table 6: Baseline comparison by subsystem.
- Figure 1: System architecture.
- Figure 2: Unsafe retrieval and contaminated context versus memory size.
- Figure 3: Cross-agent propagation depth versus agent count.
- Figure 4: Security-cost frontier.
- Figure 5: Threshold sensitivity or top-k source diversity.

Chapter 8

Metrics and Evaluation Logic

8.1 Notation

Let Q_a be attack queries, Q_b be benign queries, P be poisoned records, $R(q)$ be the final retrieved set for query q , and $C(q)$ be the final memory-conditioned context shown to the model. Let U be task utility and T be end-to-end latency.

8.2 Metric Definitions

Metric	Formal definition	Intuition and computation	Supports / weakens claim	Range
Attack Success Rate (ASR)	$\frac{1}{ Q_a } \sum_{q \in Q_a} \mathbb{1}[\text{attack successful}]$	Proportion of attacked tasks where the attacker achieves the scenario goal. Compute from grader logs.	Lower ASR supports security; unchanged ASR weakens the paper.	$[0, 1]$
Attack Success Reduction	$1 - \frac{\text{ASR}_{def}}{\text{ASR}_{base}}$	Relative reduction against no-defense or baseline.	Large positive reduction supports headline claim; negative values mean defense is harmful.	$(-\infty, 1]$
Unsafe Retrieval Rate (URR)	$\frac{1}{ Q_a } \sum_{q \in Q_a} \mathbb{1}[R(q) \cap P \neq \emptyset]$	Measures whether poisoned records remain in final retrieval. Requires retrieved IDs and poison labels.	Supports retrieval safety but is insufficient alone.	$[0, 1]$

Contaminated Context Rate (CCR)	$\frac{1}{ Q } \sum_q \mathbb{I}[\text{contaminated}(C(q))]$	Measures what the model actually sees after context shaping.	Stronger than URR; high CCR weakens context-shaping claims.	$[0, 1]$
Propagation Depth	Maximum or mean graph distance from initial poisoned write to downstream contaminated writes/actions.	Build from parent-child lineage edges.	Lower depth supports containment; depth growing with agent count weakens claim.	≥ 0
Propagation Fan-out	Count of downstream contaminated records/actions reachable from initial poison.	Compute from lineage DAG.	Lower fan-out supports shared-memory defense.	≥ 0
Quarantine Precision	$\frac{TP}{TP+FP}$	Fraction of quarantined records that were truly poisoned. Requires poison labels.	Low precision means over-blocking.	$[0, 1]$
Quarantine Recall	$\frac{TP}{TP+FN}$	Fraction of poisoned writes caught by quarantine or down-trust policy.	Low recall means poisoned writes pass through.	$[0, 1]$
Benign Utility Preservation (BUP)	$\frac{U_{def}^{clean}}{U_{base}^{clean}}$	Clean-task success ratio under defense versus baseline.	Values near 1 support deployability; large drops are fatal.	$[0, \infty)$
Task Success Under Defense (TSD)	U_{def}^{attack}	Utility on attacked tasks while defense is enabled.	Security is less meaningful if the agent stops solving tasks.	Task-specific
Latency Overhead	$\frac{T_{def} - T_{base}}{T_{base}}$ at p50 and p95	End-to-end and per-stage timing.	Moderate overhead supports practicality; high p95 weakens systems claim.	≥ -1

Memory Growth Overhead	$\frac{S_{def}-S_{base}}{S_{base}}$	Extra records/bytes from metadata, logs, quarantine, and lineage.	Supports storage scalability if bounded.	≥ -1
Scalability Slope	$\hat{\beta}$ in $Y = \alpha + \beta \log N$ for $Y \in \{\text{latency}, URR, CCR\}$	Fit slopes over memory records, agents, or top-k.	Flatter security degradation supports scaling.	Real-valued
Top-k Source Diversity	$\frac{ \text{unique sources in } R(q) }{ R(q) }$ averaged over queries	Measures whether retrieval is dominated by one source or lineage.	Higher diversity supports flooding resistance.	$[0, 1]$
Contradiction Handling Rate	Fraction of conflict trials where the trusted branch is selected, both branches are surfaced, or the agent abstains.	Requires labeled contradiction pairs.	High rate supports semantic-memory safety.	$[0, 1]$

8.3 Are New Metrics Needed?

Yes. Existing attack success and utility metrics are necessary but not enough. Unsafe Retrieval Rate is common-sense and easy to compute, but it misses context shaping: a poisoned record can be retrieved and safely rendered as untrusted evidence. Contaminated Context Rate is therefore needed. Propagation Depth and Fan-out are also needed because the report’s distinctive claim involves shared-memory containment. Scalability Slope is needed because the paper’s thesis says security must survive memory growth and team growth.

Chapter 9

Baselines and Ablations

9.1 Baseline Definitions

Baseline	Definition	Why fair
No defense	Native framework memory with no additional filtering, reranking, context shaping, or propagation control.	Necessary lower bound and deployment default.
Provenance-only	Attach source labels and trust priors but do not quarantine, rerank by risk, or block tools.	Tests whether metadata alone explains results.
Spotlighting-style wrapper	Separate retrieved content from instructions and mark provenance in prompt context.	Fair retrieval-context baseline named by the report.
Trust-threshold sanitizer	Drop or sanitize records below a trust threshold.	Strong lightweight baseline close to practical defenses.
Random quarantine	Quarantine the same number of records as MemSieve at random.	Detects whether improvement comes merely from blocking volume.
RAGForensics optional	Vector/RAG-specific traceback defense if easy to integrate.	Fair only for vector/RAG rows. Do not use for semantic/shared claims.
Task Shield optional	Tool-time task-alignment defense if easy to integrate.	Fair only for tool-action rows. Do not use for write-path claims.

9.2 Comparative Evaluation Logic

A convincing comparison is not “MemSieve beats no defense.” The paper must show:

- MemSieve beats no defense on ASR, URR, CCR, and propagation.
- MemSieve beats provenance-only, proving labels alone are insufficient.
- MemSieve beats spotlighting-style context wrapping on write-path and propagation rows.
- MemSieve beats trust-threshold sanitizer on clean utility or attack reduction because it is more selective.
- MemSieve remains competitive on latency and memory overhead.
- Non-RAG subsystems show real gains.

9.3 Comparison Tables to Include

Table	Purpose
Framework/subsystem coverage table	Show LangGraph, LlamaIndex, and CrewAI support for each memory subsystem and whether adapter coverage is full, partial, or missing.
Defense comparison table	Rows: defense settings. Columns: ASR reduction, URR, CCR, BUP, p95 latency, memory overhead. Split by subsystem if space allows.
Prior-defense comparison table	Rows: MemSieve, spotlighting, trust threshold, optional RAGForensics, optional Task Shield. Columns: supported lifecycle stages, memory subsystems, framework coverage, metrics improved.
Ablation table	Rows: full system and one-module-removed variants. Columns: ASR, CCR, propagation depth, BUP, p95 latency.
Scalability table	Rows: memory size, poison ratio, top-k, agent count. Columns: slope, confidence interval, dominant failure mode.

9.4 Ablation Interpretation

- If removing the write gate barely changes results, the write-path contribution is weak.
- If removing context shaping barely changes CCR, the context-shaping contribution is weak.
- If removing propagation control barely changes fan-out, the distinctive multi-agent claim is weak.
- If full MemSieve is only marginally better than trust-threshold sanitizer, simplify claims and emphasize portability and logging rather than defense quality.
- If clean utility drops more than 10 percent on core tasks, the paper becomes difficult to defend.

Chapter 10

Paper Skeleton

10.1 Title Candidates

- MemSieve: A Provenance-Aware Memory Firewall for LLM Agents
- Defending Mutable Memory in LLM Agents with Provenance and Lineage
- Memory Firewalls for Stateful LLM Agents
- Write, Retrieve, Propagate: Securing Agent Memory Across Frameworks

10.2 Abstract Draft Structure

1. State the problem: LLM agents increasingly rely on writable, persistent memory.
2. State the gap: existing defenses are fragmented across retrieval, prompts, or tools.
3. State the system: MemSieve is a provenance-aware memory firewall for write, retrieval, context, tool, and propagation paths.
4. State the method: common schema, quarantine, risk-aware retrieval, typed evidence, tool verification, lineage control.
5. State the evidence: three frameworks, four memory subsystems, six scenarios, baselines, ablations, scalability sweeps.
6. State the result: fill in quantitative reductions only after experiments are complete.

Missing / Must Decide: Do not write numeric claims in the abstract until final results are generated.

10.3 Paper Sections

Section	What to write	Evidence/figures	Reviewer objection to preempt
Introduction	Writable memory changes agent security from one-turn prompt defense to persistent state defense. State contributions crisply.	Motivating example and contribution list.	“This is just another RAG defense.”
Background	Define vector/RAG, semantic, episodic, shared coordination memory. Explain public framework support.	Subsystem table.	“The memory taxonomy is arbitrary.”
Threat Model	Define attacker capabilities, goals, assumptions, in/out scope.	Capability-to- scenario table.	“The attacker is too strong or vague.”
System Design	Present schema, gates, scoring, context shaping, tool verification, propagation.	Architecture figure, schema snippet, scoring equation.	“The defense is ad hoc.”
Implementation	Explain adapters and logging at high level.	Framework coverage table.	“Framework-agnostic is not implemented.”
Methodology	Describe tasks, scenarios, baselines, metrics, seeds, model parameters.	Scenario matrix, metric table.	“The evaluation is not reproducible.”
Evaluation	Report attack reduction, contaminated context, clean utility, ablations.	Main tables and ablation table.	“Security comes from over-blocking.”
Scalability	Show memory-size, poison-ratio, top-k, and agent-count sweeps.	Three scalability figures.	“It only works on toy stores.”
Discussion	Analyze failures, tradeoffs, and deployment constraints.	Error analysis examples.	“The paper hides limitations.”

Related Work	Map memory architectures, memory attacks, RAG defenses, prompt injection defenses, tool defenses, multi-agent propagation.	No figure required.	“Novelty is unclear.”
Limitations and Ethics	State synthetic setting, hosted model reproducibility, no formal guarantee, safe stub tools.	Artifact statement.	“The evaluation is unsafe or overclaimed.”
Conclusion	Restate bounded result and why lifecycle memory defense matters.	None.	“Conclusion introduces new claims.”
Appendix	Thresholds, prompts, scenario manifests, extra plots, adapter details.	Full artifact details.	“Key details omitted.”

10.4 Related Work Map

- Agent memory architectures: modular memory taxonomies, long-term memory, semantic/episodic framing, shared memory.
- Memory and RAG attacks: indirect prompt injection, PoisonedRAG, AgentPoison, MINJA, Zombie Agents, memory injection.
- Agent security benchmarks: InjecAgent, AgentDojo, Agent Security Bench.
- Prompt and retrieval defenses: spotlighting, structured queries, provenance cueing, trust-aware sanitization, RAGForensics.
- Tool and task defenses: Task Shield and task-alignment gating.
- Multi-agent propagation: prompt infection and remediation-style work.

Missing / Must Decide: Bibliographic metadata must be verified from primary sources before paper submission. The source report provides URLs and names, not complete citations.

Chapter 11

Timeline and Execution Checklist

11.1 Two-Week Timeline

Assume work starts on May 2, 2026 and the goal is a submission-quality compact paper by May 16, 2026. This timeline is aggressive and only realistic if the MVP stays narrow.

Date	Coding milestone	Writing/figure milestone	Hard rule
May 2	Freeze title, thesis, threat model, frameworks, subsystems, baselines.	Draft abstract slots and intro claims.	No code before claims are bounded.
May 3	Smoke-test LangGraph, LlamaIndex, CrewAI; define schema and logs.	Draft threat model and related-work map.	Cut any framework that does not run cleanly after a short attempt.
May 4	Implement core schema, audit logger, LangGraph adapter, write gate.	Draft architecture subsection.	Get one defended end-to-end run.
May 5	Port write path to LlamaIndex and CrewAI.	Draft framework coverage table.	Keep schema identical across frameworks.
May 6	Implement quarantine, duplicate throttling, contradiction checks.	Draft metric definitions.	No model training.
May 7	Implement retrieval reranking, source diversity, context shaping.	Draft methodology.	Produce first retrieval table.
May 8	Implement tool verifier and CrewAI propagation controller.	Draft evaluation section shell.	Fix lineage logging before running more propagation.

May 9	Run mock matrix: one model, three frameworks, two scenarios, three baselines.	Freeze version-one claims.	Do not expand scope after this point.
May 10	Add episodic scenario and flooding stress; optional AutoGen only if easy.	Start figures.	Cut AutoGen immediately if messy.
May 11	Run clean utility and conflict/recency experiments.	Fill utility and ablation tables.	Clean utility is mandatory.
May 12	Run memory-size and top-k sweeps.	Draft scalability section.	At least one scaling curve must exist.
May 13	Run agent-count propagation sweeps; optional second model if done.	Refine captions and claims.	Do not add second model before propagation.
May 14	Compute final metrics, error analysis, ablations.	Complete full paper draft.	Cut weak modules rather than hide them.
May 15	Harsh internal review, threat-model tightening, table cleanup.	Near-final submission draft.	Make claims smaller and sharper.
May 16	Reproducibility package, appendix, final compile, artifact naming.	Final submit.	No last-minute new experiments.

11.2 Minimum Viable Paper Path

- Three frameworks: LangGraph, LlamaIndex, CrewAI.
- Four subsystems: vector/RAG, semantic, episodic, shared coordination.
- Five defense settings: no defense, provenance-only, spotlighting, trust threshold, full Mem-Sieve.
- Six scenarios: poisoned fact, retrieved instruction, experience graft, conflict/recency, shared cascade, flooding.
- One primary model and one embedding backend.
- Clean utility table, main attack table, ablation table, scalability figures.

11.3 Strong Version Path

Add only after the MVP tables are complete:

- AutoGen as a fourth framework.

- Second model transfer.
- RAGForensics on vector rows.
- Task Shield on tool rows.
- Letta/MemGPT archival/shared-memory case if integration is trivial.
- Public benchmark subset only if it plugs in cleanly on day one.

11.4 Phase Checklist

- Code scaffold complete.
- Three framework adapters smoke-tested.
- Canonical schema stable.
- Audit logging complete.
- Quarantine isolation tested.
- Retrieval reranking logged term-by-term.
- Context shaping prevents raw instruction rendering.
- Tool verifier tested with high-impact stub tools.
- Propagation lineage graph generated.
- Baselines implemented fairly.
- Scenarios auto-gradable.
- Aggregation reproduces tables from logs.
- Figures generated from scripts.

Chapter 12

Risks, Reviewer Objections, and Mitigations

12.1 Reviewer 1: Security and System Design Reviewer

Major concerns.

- Threat model may be too broad if write, retrieval, tool, and propagation attacks are all claimed without tight boundaries.
- Defense may look heuristic unless each score has a clear purpose and ablations.
- Quarantine and trust scoring may create a false sense of security without strong audit and isolation guarantees.
- Framework adapters may be shallow wrappers that do not capture real memory flows.

Minor concerns.

- Naming “firewall” may imply stronger guarantees than delivered.
- Tool verifier scope may be underspecified.
- Procedural memory is mentioned but not defended in the MVP.

Missing evidence.

- Quarantine precision/recall.
- Proof that quarantined records cannot enter normal retrieval.
- Per-stage ablation results.
- Adapter coverage table.

Weak claims. “Framework-agnostic,” “general defense,” and “memory firewall” are weak unless bounded by implementation evidence.

Required fixes. Freeze the threat model, show exact trust boundaries, log every decision, and include ablations for write gate, retrieval gate, context shaping, tool verifier, and propagation.

Borderline outcome. The paper has a nice architecture and some vector/RAG results, but shared-memory propagation is toy-like and clean utility is weak.

Strong accept outcome. The paper demonstrates real adapters, strong non-RAG results, capped propagation under agent-count scaling, bounded overhead, and honest limitations.

12.2 Reviewer 2: Machine Learning and LLM Evaluation Reviewer

Major concerns.

- Evaluation may rely on one model and synthetic tasks, limiting generality.
- Auto-graders may be brittle or biased toward the defense.
- Results may be sensitive to thresholds.
- Hosted model nondeterminism may weaken reproducibility.

Minor concerns.

- The report does not specify model, temperature, embedding backend, or prompt templates.
- Three seeds may be thin if variance is high.
- Utility metrics may be task-specific and not comparable across frameworks.

Missing evidence.

- Threshold sensitivity plot.
- Confidence intervals.
- Clean utility table.
- Second-model transfer if time allows.
- Grader validation examples.

Weak claims. Any claim that MemSieve is model-general is weak without at least a second model or a careful limitation.

Required fixes. Pin model parameters, log prompts and outputs, keep graders deterministic, run threshold sweeps, and report variance.

Borderline outcome. The system reduces attack success but only with aggressive thresholds that harm benign tasks.

Strong accept outcome. The system preserves clean utility, improvements are stable across threshold ranges, and a second model shows the main trend persists.

12.3 Reviewer 3: Conference Acceptance Reviewer

Major concerns.

- Novelty may be unclear in a crowded prompt-injection and RAG-defense space.
- The paper may be too broad for its evidence.
- Baseline comparison may be incomplete or unfair.
- The work may read as engineering middleware rather than research.

Minor concerns.

- Related work may be under-verified because the source report gives URLs, not full citations.
- The title may overpromise if results are modest.
- Page budget may be tight for architecture, methodology, and evaluation.

Missing evidence.

- A single table showing cross-framework and cross-subsystem results.
- A clear comparison with simple and strong baselines.
- Scalability curves that tie security outcome to scale.
- Error analysis explaining failure cases.

Weak claims. “First,” “comprehensive,” and “general” should be avoided unless verified against prior work.

Required fixes. Make the contribution sentence brutally precise: lifecycle memory defense across four memory subsystems and three public frameworks, evaluated under scaling pressure.

Borderline outcome. The paper is coherent but looks like a collection of wrappers with small synthetic experiments.

Strong accept outcome. The paper changes how reviewers think about agent memory security by showing that write, retrieve, and propagate paths must be evaluated together.

12.4 Global Risk Table

Risk	Severity	Why serious	Mitigation
Looks like another RAG defense	High	Most public defenses already target retrieval.	Put semantic, episodic, and shared-memory results in the main table.
Threat model vague	High	Security reviewers punish undefined attacker powers.	Capability-to- scenario table and strict out-of-scope list.
Defense too heuristic	High	Ad hoc filters are weak novelty.	Interpretable feature terms, threshold sweeps, and ablations.
Experiments too small	High	Cross-framework claim needs real breadth.	Three frameworks, four subsystems, multiple seeds, scaling sweeps.
Clean utility loss	High	Over-blocking is not a useful defense.	BUP table and security-cost frontier.
Framework integration failure	Medium to high	Adapter claim collapses.	Cut optional frameworks early; keep MVP adapters simple.
Scalability synthetic only	Medium to high	Reviewers may dismiss toy scaling.	Use real runtime paths and real framework stores where possible.
Baseline reproduction sinkhole	Medium	Reproducing too many defenses wastes time.	Include three clean baselines; optional prior defenses only when easy and fair.

12.5 Final Deliverable Checklist

Before submission, every item below must be true:

- Code complete for core schema, gates, stores, adapters, baselines, scenarios, runner, metrics, aggregation, and plotting.
- LangGraph, LlamaIndex, and CrewAI adapters smoke-tested and used in main results.
- Experiments complete for clean, attacked, defended, and ablation conditions.
- Tables generated from raw logs: framework coverage, scenario matrix, main attack results, clean utility, ablations, baseline comparison, scalability summary.
- Figures generated from scripts: architecture, lineage graph, memory-size scaling, agent-count propagation, security-cost frontier, threshold sensitivity if space allows.

- Logs saved for every run: manifests, records, decisions, retrievals, contexts, tool calls, lineage, grader outputs, timings.
- Claims validated against evidence; no abstract numeric claim appears without a table or figure.
- Related work checked from primary sources and full citations completed.
- Threat model bounded and out-of-scope list included.
- Clean utility loss reported honestly.
- Reviewer objections addressed in introduction, methodology, evaluation, and limitations.
- Paper compiles from a clean checkout.
- Artifact README explains how to reproduce every table and figure.
- Comparison table included to show whether MemSieve outperforms no defense, provenance-only, spotlighting-style wrapper, trust-threshold sanitizer, random quarantine, and optional prior defenses where applicable.

12.6 Harsh Final Verdict

The best version of this project is narrow, disciplined, and empirical. Build the smallest system that undeniably shows cross-subsystem memory defense plus scaling. The fastest way to weaken the paper is to overreach on frameworks, domains, baseline reproductions, or claims of generality. The fastest way to make it publishable is to keep the claim bounded, the runtime pipeline crisp, the logs complete, the baselines fair, and the scalability evidence central.

Bibliography

- [1] `paper_roadmap.pdf`, repository source report, extracted May 2, 2026.
- [2] Source report cited CoALA-style modular agent memory taxonomies: <https://arxiv.org/abs/2309.02427>.
- [3] Source report cited RAGForensics: <https://arxiv.org/abs/2504.21668>.
- [4] Source report cited public memory poisoning work: <https://arxiv.org/abs/2402.07867>.
- [5] Source report cited LangGraph/LangChain memory documentation: <https://docs.langchain.com/oss/javascript/concepts/memory>.
- [6] Source report cited indirect prompt injection: <https://arxiv.org/abs/2302.12173>.
- [7] Source report cited prompt infection / multi-agent propagation: <https://arxiv.org/abs/2410.07283>.
- [8] Source report cited spotlighting: <https://arxiv.org/abs/2403.14720>.
- [9] Source report cited agent security benchmarks: <https://arxiv.org/abs/2403.02691>.
- [10] Source report cited LlamaIndex memory documentation: https://docs.llamaindex.ai/en/stable/module_guides/deploying/agents/memory/.
- [11] Source report cited CrewAI memory documentation: <https://docs.crewai.com/en/concepts/memory>.
- [12] Source report cited AutoGen memory documentation: <https://microsoft.github.io/autogen/dev/user-guide/agentchat-user-guide/memory.html>.
- [13] Source report cited MemGPT/Letta memory hierarchy: <https://arxiv.org/abs/2310.08560>.
- [14] Source report cited trust-aware sanitization: <https://arxiv.org/abs/2601.05504>.